Magnetic Levitation Project Proposal

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1 Project overview

Our project will suspend a permanent magnet carrying an object below an electromagnet. The magnet, carrier, and object will total 10 g. The webpage will command motion between 10 and 20 mm surface gaps. Two Hall sensors, mounted above and below the electromagnet, will measure magnetic field. We will combine both readings with measured coil current to estimate the target's height. A position PI controller will adjust PWM duty to hold the requested gap; there will be no current-feedback controller.

We will use the STM32L432KC MCU, UPduino FPGA, and course ESP8266 Wi-Fi module. Gap d is the distance from the bottom of the coil to the top of the suspended magnet. We will demonstrate the round trips and position tolerance listed in the specifications and check gap with a ruler. The webpage will display requested gap, estimated gap, current, and faults on a real-time graph, include a stop button, and allow updates to the position PI constants.

2 MCU and FPGA design

The MCU will read the MCP3204 ADC over SPI using three separate channels: Hall sensor A, Hall sensor B, and coil current. A timer will start scans at 5 kHz, with DMA moving SPI data. At 1 kHz, the MCU will estimate gap from the calibrated sensor readings and measured current. USART1 will connect to the ESP8266 at 125000 baud. The MCU will handle the web interface, telemetry, and transfer of requested gap and PI constants.

The MCU will send estimated gap, requested gap, measured current, and status to the FPGA over SPI. The FPGA will execute the position PI controller at 1 kHz using DSP blocks and generate 20 kHz PWM. Its output is duty directly; there is no requested-current signal or current PI loop. Current measurement supports magnetic-field compensation, telemetry, and protection. The FPGA will turn off drive after 50 ms without valid commands. A MOSFET, gate driver, and flyback diode will switch the 12 V coil; a shunt and amplifier will measure current. An E-stop and overcurrent comparator will disable the driver directly. The MCU will also provide a drive-permit signal.

SPI DMA is our new MCU peripheral use; the DSP position controller is our new FPGA logic; the ADC and magnetic sensing circuit are our new external hardware. SPI links carry commands and replies. Logic uses 3.3 V and shares ground with the coil circuit. Figure 1 shows the updated architecture. PI gain changes will be range-checked and applied together at a control-update boundary. Drive limits and integrator anti-windup will be implemented and tested.

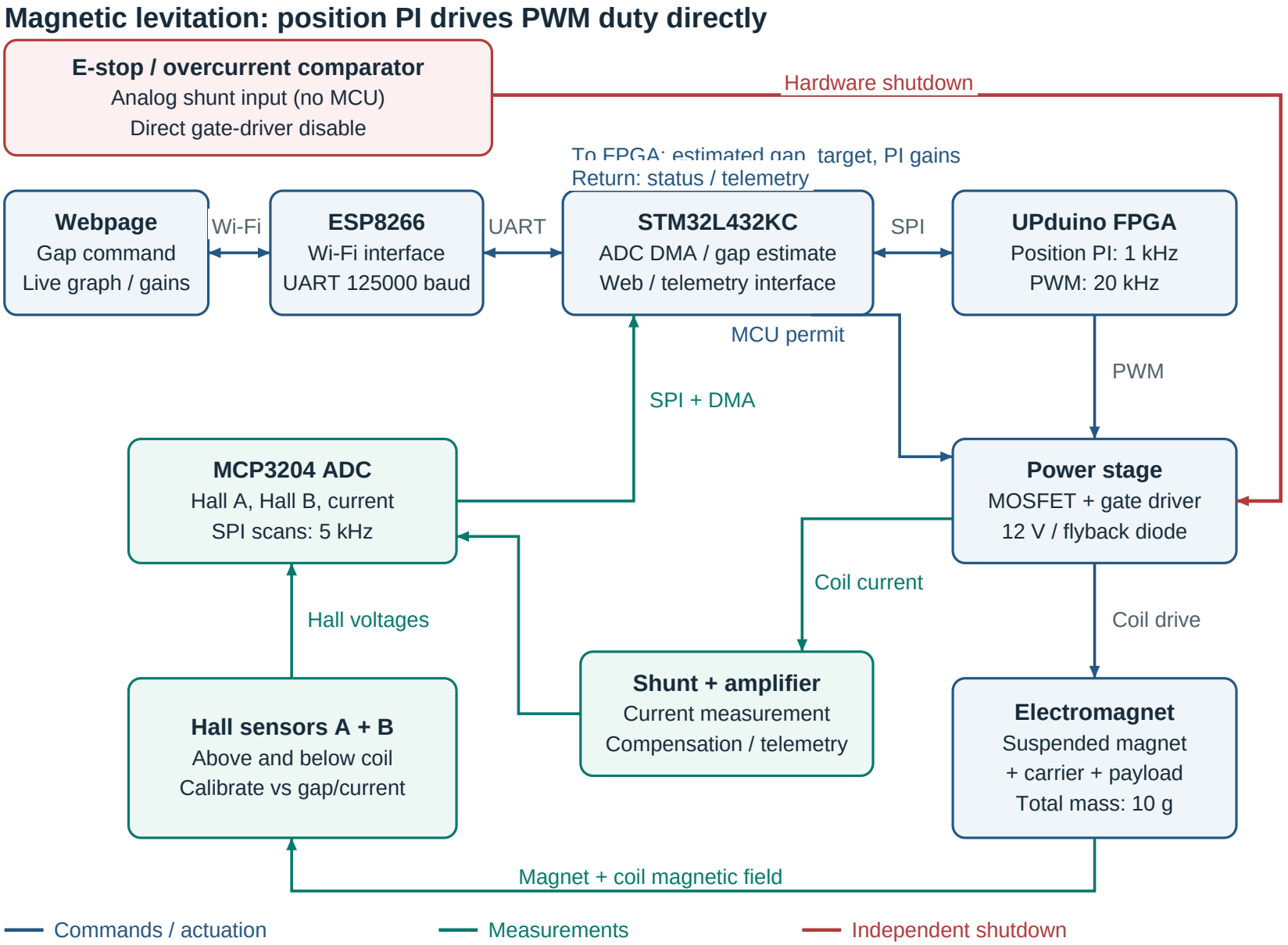


Figure 1: FPGA position PI drives PWM duty. The MCU handles ADC/DMA, gap estimation, and the web interface. The analog overcurrent comparator disables drive independently of the MCU. There is no current controller.

3 Biggest challenge

Our biggest challenge is separating the target's field from the coil's field. We will mount Hall sensors A and B above and below the electromagnet and calibrate both readings at known gaps and coil currents on a plastic fixture. The MCU will read the sensors on separate ADC channels, combine their readings in software, and compensate for the coil field using measured current. We will verify the estimated gap against ruler measurements over the 10–20 mm operating range. The ADC's limited differential mode cannot directly subtract both full Hall outputs. A catch tray and upper stop will protect against falls and the target snapping upward.

4 Current needed to lift an object at 12 V

Voltage alone does not tell us how much weight we can lift. We need the total mass m of the magnet, carrier, and object, plus the coil geometry and magnet strength. For a first estimate, assume a small aligned permanent magnet below a thin air-core coil. Let N be the number of turns, a the mean coil radius, μ the magnet's dipole moment, and z the distance from the coil plane to the magnet's center. Use meters and kilograms in the equations.

$$B(z) = \frac{\mu_0 N I a^2}{2(a^2 + z^2)^{3/2}}, \tag{1}$$

$$F \simeq \mu \left| \frac{\mathrm{d}B}{\mathrm{d}z} \right| = K(z) I, \tag{2}$$

$$K(z) = \frac{3\mu_0 N \mu a^2 z}{2(a^2 + z^2)^{5/2}}, \qquad I_{\mathrm{hold}}(z) = \frac{mg}{K(z)}. \tag{3}$$

Here $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H/m}$ and $g = 9.81\,\mathrm{m/s^2}$. To move up by Δh mm from z_0, the final distance is $z_1 = z_0 - \Delta h/1000$. Current must provide enough force throughout the move, not just at the endpoint. To accelerate upward at a_{up}, use $I = m(g + a_{\mathrm{up}})/K(z)$. Center distance z differs from surface gap d; we will convert between them using the actual dimensions.

4.1 Example for a 10 g magnet and payload

For a sizing calculation, use $N = 500$, $a = 10\,\mathrm{mm}$, $\mu = 0.050\,\mathrm{A\,m^2}$, and $R = 8\,\Omega$. These are calculation assumptions, not selected parts. The table covers center distances of 10–20 mm; actual center distances include the coil and magnet offsets from surface gap d. With ideal PWM, $D \simeq IR/12$ and coil heating is $P = I^2 R$.

Center distance z	Holding current	12 V duty	Coil heating
20 mm	5.819 A	388%	270.9 W
15 mm	2.643 A	176%	55.9 W
10 mm	1.178 A	79%	11.1 W

For this example, $12/8 = 1.50\,\mathrm{A}$ is the ideal full-on current limit. Holding at $z = 20\,\mathrm{mm}$ requires 5.819 A, so this example coil cannot deliver the required range at 12 V. A 10 mm upward move from $z = 20$ to 10 mm requires the largest holding current at the start. For a one-second move with equal acceleration and deceleration, $a_{\mathrm{up}} = 4(0.010)/1^2 = 0.040\,\mathrm{m/s^2}$; the initial current becomes 5.842 A. A duty above 100% means the supply cannot provide the required current.

We will select the final coil and magnet using measured force versus gap and current. The assembly must support the 10 g target over the full 10–20 mm surface-gap range at 12 V. We will measure hot resistance, inductance, and temperature and check that the supply covers $IR + L\,\mathrm{d}I/\mathrm{d}t$ and circuit losses. An iron core and finite magnet dimensions change the simple air-core estimate. Stable suspension will be verified by the demonstration specifications.

5 Timing and measurement calculations

At 1 MHz SPI, a 24-clock ADC reading takes 24 μs. Including a 1 μs gap, current plus two Hall readings takes 75 μs. An 8-byte FPGA exchange takes about 68 μs with chip-select gaps. Together this uses 143 μs of a 200 μs scan period, leaving 57 μs for setup and processing. We will check the actual timing with a logic analyzer. The FPGA's position PI update is 1 kHz; ADC acquisition and SPI exchanges may run at the faster 5 kHz scan rate.

The 12-bit ADC has 4096 levels: $3.3\,\mathrm{V}/4096 = 0.806\,\mathrm{mV}$ per count. With a $0.100\,\Omega$ shunt and gain of 20, current resolution is about 0.403 mA per count. At 1.18 A the shunt dissipates 0.139 W. A 24 MHz FPGA clock divided by 1200 gives 20 kHz PWM.

6 Stability calculation: nominal 10 mm operating point

The controller is position PI driving duty directly, with no current controller. For a real 10 mm surface gap d_0, the force model must use $z_0 = d_0 + \Delta_g$, where Δ_g accounts for coil-plane and magnet-center offsets. The numerical illustration below uses $z_0 = 10\,\mathrm{mm}$ in the thin-coil/point-magnet approximation; it is not a validated result for a finite assembly at 10 mm surface gap.

For $m = 0.010\,\mathrm{kg}$, the equilibrium force is $F_0 = mg = 0.0981\,\mathrm{N}$. Using the sizing model above gives

$$k_f = K(z_0) = 0.0833041\,\mathrm{N/A}, \qquad i_0 = \frac{mg}{k_f} = 1.17761\,\mathrm{A}, \qquad u_0 = \frac{Ri_0}{12} = 0.78508. \tag{4}$$

Define $x = d - d_0$, $j = i - i_0$, and upward force $F \simeq mg - k_x x + k_f j$. The local destabilizing force slope is

$$k_x = -\left.\frac{\partial F}{\partial d}\right|_0 = mg\left(\frac{5z_0}{a^2 + z_0^2} - \frac{1}{z_0}\right) = 14.715\,\mathrm{N/m}. \tag{5}$$

With downward displacement positive, the mechanical equation, position PI, and electrical equation are

$$m\ddot{x} + c\dot{x} - k_x x + k_f j = 0, \tag{6}$$

$$\delta u = K_p x + K_i \eta, \qquad \dot{\eta} = x, \tag{7}$$

$$L\dot{j} + Rj - k_f \dot{x} = V_s \delta u. \tag{8}$$

Here c is passive mechanical damping, L is differential coil inductance, $\delta u = u - u_0$, and $V_s = 12\,\mathrm{V}$. The PI gains have units m^{-1} and $\mathrm{m}^{-1}\mathrm{s}^{-1}$, respectively. The induced-voltage term follows reciprocal magnetic coupling; it is not a current controller.

Taking Laplace transforms and eliminating current gives the characteristic equation

$$\boxed{\begin{aligned} 0 = {} & mLs^4 + (mR + cL)s^3 + (cR - k_x L + k_f^2)s^2 \\ & + (k_f V_s K_p - k_x R)s + k_f V_s K_i. \end{aligned}} \tag{9}$$

For the explicitly assumed $L = 10\,\mathrm{mH}$ and negligible passive damping $c = 0$, this becomes

$$\boxed{\begin{aligned} 0 = {} & 0.0001s^4 + 0.08s^3 - 0.140210s^2 \\ & + (0.999649K_p - 117.72)s + 0.999649K_i. \end{aligned}} \tag{10}$$

The negative s^2 coefficient violates a necessary Routh–Hurwitz condition. Therefore this assumed continuous-time plant cannot be stabilized by position PI tuning alone. For illustration, $K_p = 150\,\mathrm{m^{-1}}$, $K_i = 100\,\mathrm{m^{-1}s^{-1}}$ produce poles

$$s = -802.247, \quad -2.9963, \quad 2.6214 \pm j\,20.2237\ \mathrm{s^{-1}}, \tag{11}$$

so an oscillatory mode grows. These are unstable example gains, not recommended settings. A necessary damping/electrical condition is

$$cR + k_f^2 > k_x L, \tag{12}$$

which requires $L < 0.4716\,\mathrm{mH}$ if $c = 0$, or $c > 0.01753\,\mathrm{N\,s/m}$ if $L = 10\,\mathrm{mH}$. Passing this condition alone does not guarantee stability.

For completeness, if the quartic coefficients are $A_4, \ldots, A_0$, all must be positive, with $A_3A_2 > A_4A_1$ and $A_3A_2A_1 > A_4A_1^2 + A_3^2A_0$. We will measure the final coil's force slopes, resistance, inductance, and damping at the actual surface gap before selecting gains. The 1 kHz sampled controller, held PWM duty, sensor filtering, and delays must then be checked for poles inside the unit circle. The companion `calculation_summary.tex` provides the derivation and sampled-state formulation. The 10 mm distance improves static lifting feasibility; it does not establish stable suspension.

7 Position resolution and stability verification

Position resolution will be determined from the calibrated relationship between the two Hall readings, measured current, and gap. ADC voltage resolution alone is not a position-resolution specification. We will measure estimator noise and bias at fixed known gaps, then report the smallest reliably distinguishable position change.

We will evaluate stability separately by recording requested and estimated gap during holds, commanded moves, and repeatable disturbances. Position band and recovery time will be stated separately. The ruler-based demonstration band below is $\pm 5\,\mathrm{mm}$; finer claims require correspondingly accurate position measurements. The calculation above identifies a design condition to resolve before stable suspension can be claimed.

8 Project specifications

These are proposed demonstration targets. We will use one magnet, carrier, and object totaling 10 g. Professors can check each box during the demonstration using a ruler, stopwatch, webpage, and scope or logic analyzer. Excellence includes all proficiency checks.

8.1 Hardware and levitation

Proficiency

☐ Use the webpage to move from 20 to 10 mm and back five times without contact.

Excellence

☐ Complete five 20-to-10 mm round trips, reaching within ±5 mm of each requested gap.

8.2 Control and webpage

Proficiency

☐ Show requested gap, estimated gap, coil current, and fault state on a real-time webpage graph, with no visible latency between the object and graph during the demonstration.
☐ Allow updates to the position PI constants from the webpage.

Excellence

☐ Demonstrate visibly noticeable movement of the magnet through webpage control across the controllable range.

9 Bill of materials and budget

Parts	Planned cost
NUCLEO-L432KC, UPduino, ESP8266; bench supply and frame	Course kit / stockroom*
2 DRV5055 Hall sensors; MCP3204 ADC; INA240A1 amplifier	$16
0.100 Ω, 1 W shunt; UCC27511 driver; MOSFET; flyback diode	$6
Coil, magnet and carrier; adapters, resistors and capacitors	$14
Fuse and connectors*; shipping and tax allowance	$3 + $10
Estimated total	**$49**

*Stockroom availability is unconfirmed. Costs are allowances; final coil and power-switch parts depend on testing.

10 Schedule and division of work

Week	Stephen	Ben
1–2	Measure coil force; build driver and FPGA PWM	Read ADC; calibrate Hall and current readings
3–4	Implement and tune FPGA position PI after plant characterization; build frame and stops	Validate gap estimate; verify loop timing and position measurements
5–6	Webpage, shutdown tests, and schematic	Motion tests, checkoff measurements, and report

Calculation references

1. OpenStax, *University Physics, Volume 2*, Sec. 12.4: Magnetic Field of a Current Loop.
2. H. H. Woodson and J. R. Melcher, *Electromechanical Dynamics*, Part I, Chs. 3 and 5. MIT OpenCourseWare. Online textbook and chapters.
3. University of Illinois, ECE 486 Handbook, Lecture 7, Routh–Hurwitz criterion. Online lecture notes.
4. A. V. Oppenheim, A. S. Willsky, with S. H. Nawab, *Signals and Systems*, 2nd ed., Chs. 9–11. MIT reading guide.